

*** Eq^m of 2 force Members: Let $\vec{P}$ & $\vec{Q}$ be the forces.
 For eq^m: $\sum \vec{F} = 0 \Rightarrow \vec{P} + \vec{Q} = 0$
 $\Rightarrow \vec{P} = -\vec{Q}$ 1.: both forces have same mag.

$$\sum \vec{M} = 0$$

2.: opposite dirⁿ.

The only kind of Moment that can exist is couple.

3.: Both forces have same line of action.

Note: If a slender rod is a 2 force Member in eq^m.

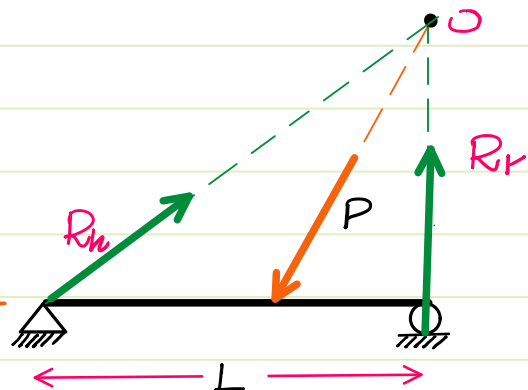
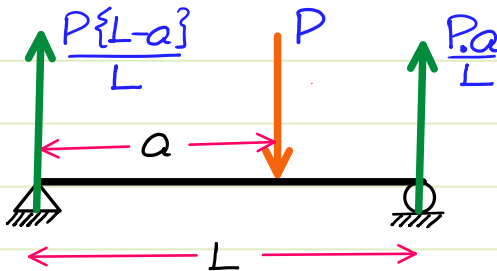
Then the rod must exp. only Axial load.



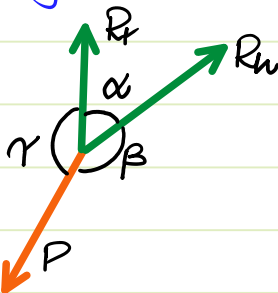
*** Eq^m of 3 force Members: The forces must be Coplanar.

A. forces may be Parallel.

B. forces are concurrent.



By Pr. of Transmissibility



$$\text{For Eq^m: } \sum \vec{M}_O = 0$$

$$\Rightarrow \cancel{M_O^P} + \cancel{M_O^{R_h}} + M_O^{R_v} = 0 \Rightarrow M_O^{R_v} = 0$$

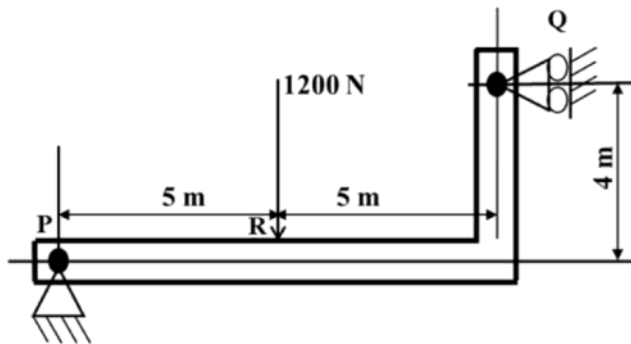
$\therefore$ L.O.A. of $\vec{R}_h$ must pass thru O

$\therefore$ The forces are concurrent.

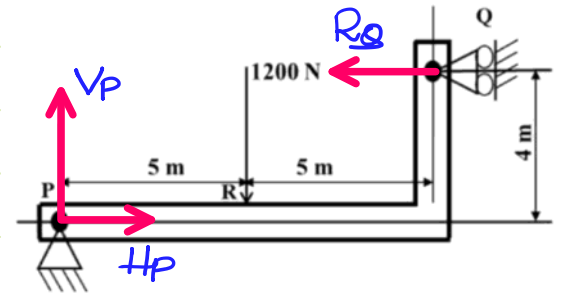
$$\text{Lami's Th^m: } \frac{P}{\sin \alpha} = \frac{R_r}{\sin \beta} = \frac{R_h}{\sin \gamma}$$

GATE IM.

A beam of negligible mass is hinged at support P and has a roller support Q as shown in the figure.



A point load of 1200 N is applied at point R. The magnitude of the reaction force at support Q is _____ N.



No. of unknowns : 3 { H_p, V_p, R_Q }

To find : R_Q

Observe the other 2 forces.
{ V_p & H_p }

* If the other 2 forces are $\parallel^l$: $\sum F_{\perp}^{or} axis = 0$

* If the other 2 forces are conc : $\sum M_{abt conc} = 0$

$$\sum M_P = 0 \Rightarrow -1200 \{5\} + R_Q \{4\} = 0$$

$$\therefore R_Q = 1500 \text{ N.}$$

Alternate Method : If a system has a total of 4 forces, such that there are 2 pairs of $\parallel^l$ forces.

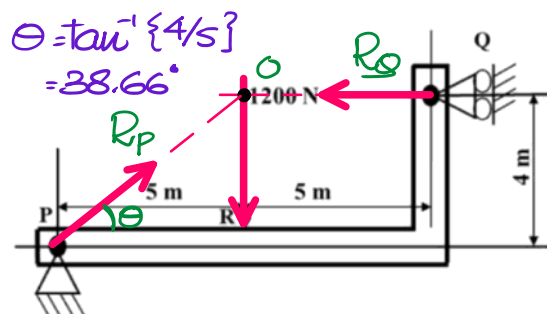
Couple & Counter couple.

1200 & V_p form a Couple & H_p & R_Q form counter couple

$$1200 \{5\} = R_Q \{4\}$$

$$R_Q = 1500 \text{ N.}$$

Alternate Method : If the No. of forces can be reduced to 3, then apply Lami's th^{rm}.



$$\theta = \tan^{-1} \{4/5\}$$

$$= 38.66^\circ$$

By Lami's

$$\frac{1200}{\sin(180^\circ - \theta)} = \frac{R_Q}{\sin(90^\circ + \theta)}$$

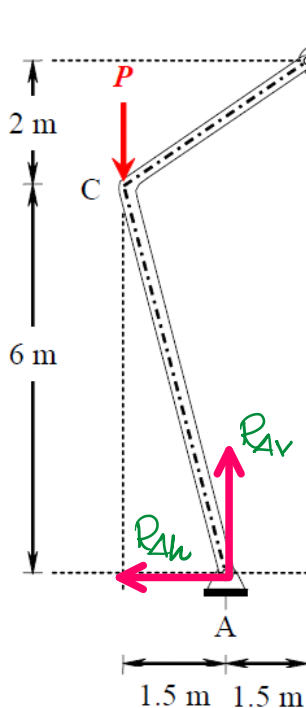
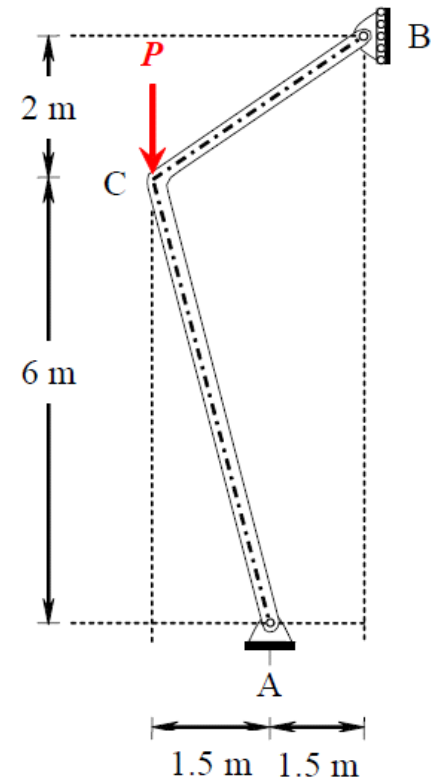
$$R_Q = \frac{1200 \cos \theta}{\sin \theta}$$

$$R_Q = \frac{1200}{\tan \theta} = 1500 \text{ N.}$$

2M CE.

A rigid member ACB is shown in the figure. The member is supported at A and B by pinned and guided roller supports, respectively. A force P acts at C as shown. Let R_{Ah} and R_{Bh} be the horizontal reactions at supports A and B, respectively, and R_{Av} be the vertical reaction at support A. Selfweight of the member may be ignored. Which one of the following sets gives the correct magnitudes of R_{Av} , R_{Bh} and R_{Ah} ?

- (A) $R_{Av} = 0$; $R_{Bh} = \frac{1}{3}P$; and $R_{Ah} = \frac{2}{3}P$
- (B) $R_{Av} = 0$; $R_{Bh} = \frac{2}{3}P$; and $R_{Ah} = \frac{1}{3}P$
- (C) $R_{Av} = P$; $R_{Bh} = \frac{3}{8}P$; and $R_{Ah} = \frac{1.5}{8}P$
- (D) $R_{Av} = P$; $R_{Bh} = \frac{1.5}{8}P$; and $R_{Ah} = \frac{1.5}{8}P$



P & R_{Av} create couple
 & R_{Ah} & R_{Bh} create counter couple.
 $\therefore |R_{Ah}| = |R_{Bh}|$

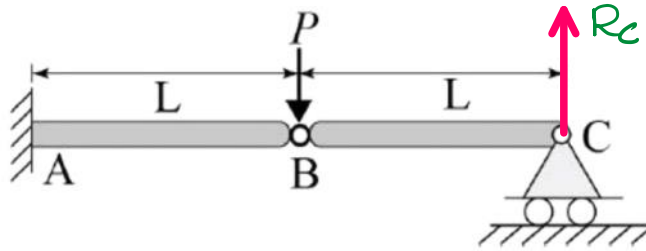
$$P\{1.5\} = R_{Ah}\{8\}$$

$$\therefore R_{Ah} = \frac{1.5P}{8} = R_{Bh}$$

$$\text{Ily } R_{Av} = P$$

2M.

The beam shown has an internal hinge at **B**. A vertical load **P = 25kN** is applied at **B**. Use **L = 2m**. Magnitude of the reactions (i.e. forces and moments) at **A** and **C** are



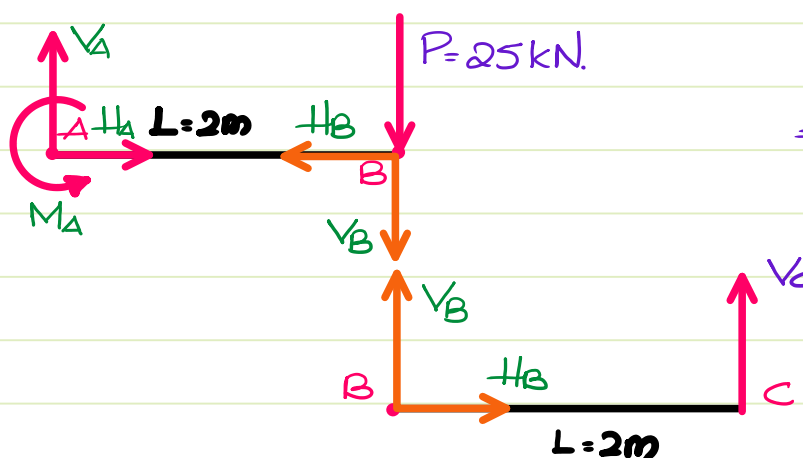
BC is a 2 force Member.
one at B & other at C.
∴ Loading Must be axial.
∴ $R_c = 0$

(A) Vertical reaction force at **C** is 12.5kN, vertical reaction force at **A** is 12.5kN, moment reaction at **A** is 0kN-m

(B) Vertical reaction force at **C** is 0kN, vertical reaction force at **A** is 25kN, moment reaction at **A** is 50kN-m

(C) Vertical reaction force at **C** is 25kN, vertical reaction force at **A** is 0kN, moment reaction at **A** is 50kN-m

(D) Vertical reaction force at **C** is 0kN, vertical reaction force at **A** is 25kN, moment reaction at **A** is 25kN-m



BC:

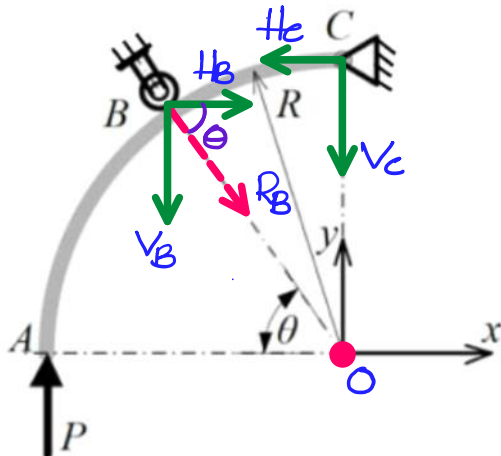
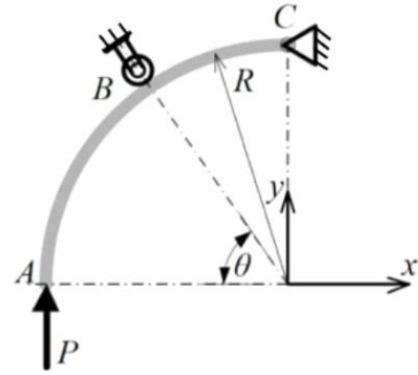
$$\begin{aligned}\sum M_B &= 0 \\ \Rightarrow V_c \{2\} &= 0 \\ \Rightarrow V_c &= 0 \\ \therefore V_b &= 0 \\ \therefore H_b &= 0\end{aligned}$$

AB:

$$\begin{aligned}\sum F_y &= 0 \Rightarrow V_A - P = 0 \\ \therefore P &= 25 \text{ kN.} \\ \sum M_A &= 0 \Rightarrow M_A - 2P = 0 \\ \therefore M_A &= 2 \times 25 \\ &= 50 \text{ kN.m.}\end{aligned}$$

2M.

A rigid rod ABC, in the form of a quarter-circular arc of radius R , is hinged at C and supported by a roller at B. A vertical force P is applied at the end A of the bar. For the reactions at B and C to be equal in magnitude, the value of the angle θ (in degrees) is (up to two decimal places).



The roller reacⁿ is similar to Normal.

$$\sum F_x = 0 \Rightarrow H_B = H_C$$

$$\because |R_B| = |R_C| \text{ \& } H_B = H_C$$

$$\therefore |V_B| = |V_C|$$

$$\sum F_y = 0 \Rightarrow P = V_B + V_C = 2V_B = 2V_C$$

$$\therefore V_B = V_C = P/2$$

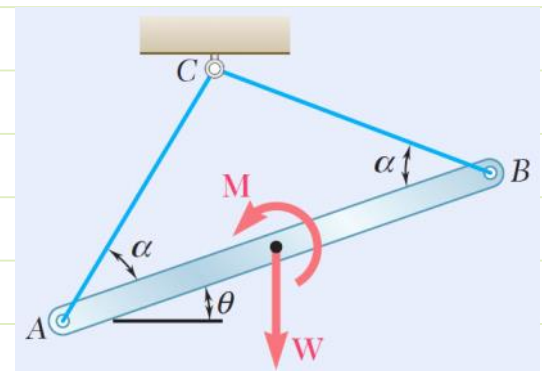
$$\sum M_O = 0 \Rightarrow -P\{R\} + H_C\{R\} = 0$$

$$\therefore H_C = H_B = P$$

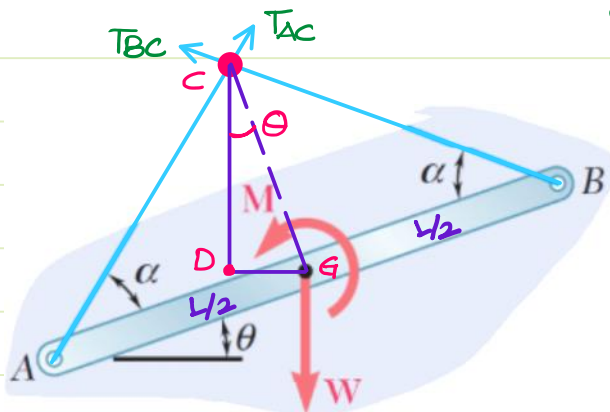
$$\theta = \tan^{-1} \left\{ \frac{V_B}{H_B} \right\} = \tan^{-1} \left\{ \frac{P/2}{P} \right\}$$

$$\therefore \theta = 26.565^\circ$$

A uniform rod AB with length L and weight W is suspended from two cords AC and BC of equal length. Derive an expression for the magnitude of the couple M required to maintain equilibrium of the rod in the position shown.



$M = f(W, L, \theta, \alpha)$ difficulty of this Q. is in Geometry.



$$\text{for eq}^m: \sum M_C = 0$$

$$\Rightarrow M - W\{DG\} = 0$$

$$\therefore M = W \cdot (DG)$$

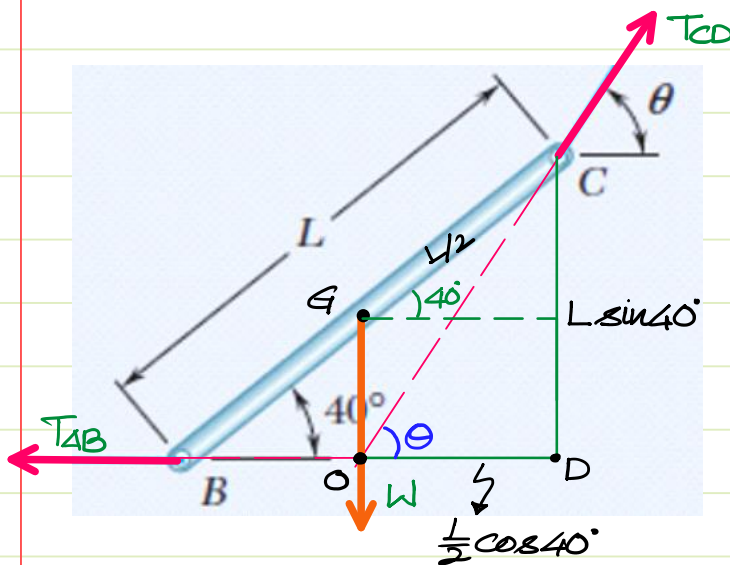
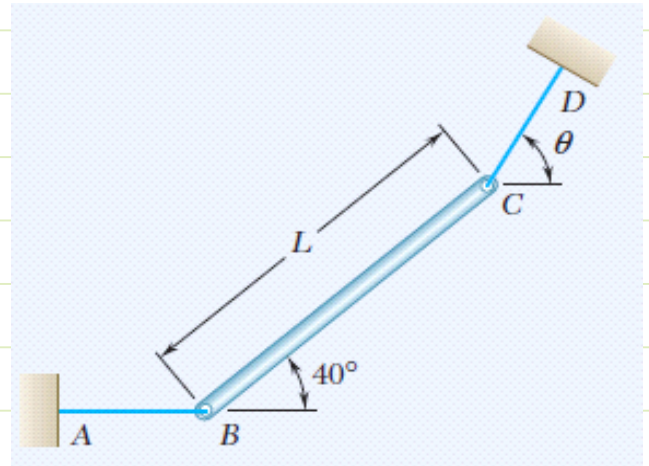
$$\Delta AGC: \tan \alpha = \frac{CG}{L/2} \Rightarrow CG = \frac{L}{2} \tan \alpha$$

$$\Delta CGD: \sin \theta = \frac{DG}{CG} \Rightarrow DG = CG \cdot \sin \theta = \frac{L}{2} \cdot \tan \alpha \cdot \sin \theta$$

$$M = \frac{W \cdot L}{2} \cdot \tan \alpha \cdot \sin \theta$$

A slender rod BC with a length of L and weight W is held by two cables as shown. Knowing that cable AB is horizontal and that the rod forms an angle of 40° with the horizontal, **determine**

- the angle θ that cable CD forms with the horizontal,
- the tension in each cable.



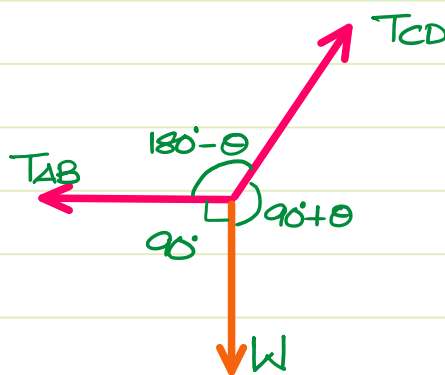
$$\theta = \tan^{-1} \left\{ \frac{CD}{OD} \right\}$$

$$\theta = \tan^{-1} \left\{ \frac{L \sin 40^\circ}{\frac{1}{2} L \cos 40^\circ} \right\}$$

$$\Rightarrow \theta = \tan^{-1} \{ 2 \tan 40^\circ \}$$

$$\theta = 59.21^\circ$$

By Principle of Transmissibility :

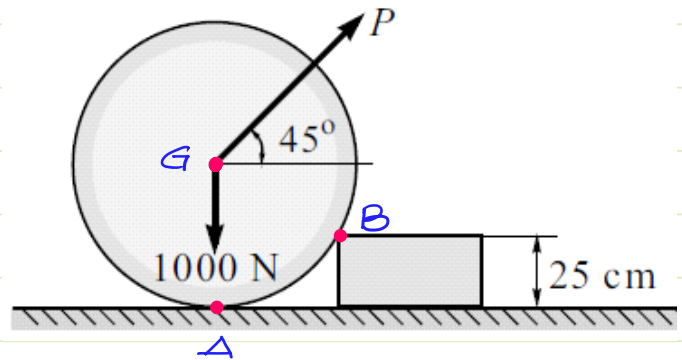


$$\frac{W}{\sin \{180^\circ - \theta\}} = \frac{T_{AB}}{\sin (90^\circ + \theta)} = \frac{T_{CD}}{\sin 90^\circ}$$

$$T_{AB} = 0.5958W$$

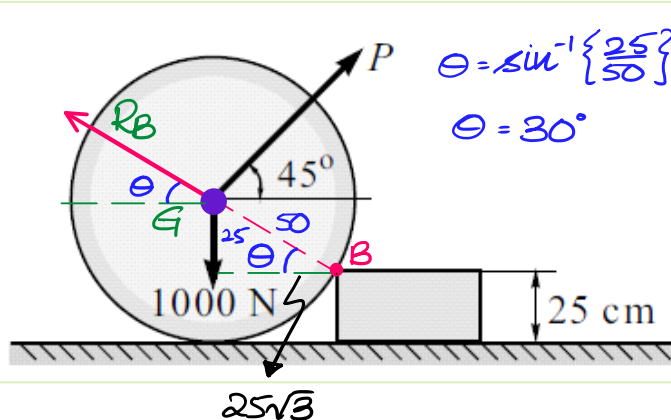
$$T_{CD} = 1.164W$$

Find the value of force **P** to impend toppling of the cylinder (**Radius = 50cm**) about the step shown. Assume all surfaces to be frictionless



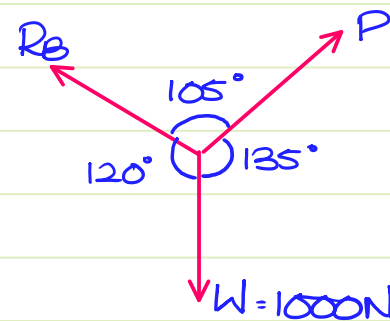
When the cylinder impends to topple abt B, the contact at A is lost. $\{N_A = 0\}$ The cylinder is now a 3 force Member $\{W, P, R_B\}$

The 3 forces must be concurrent at G.



$$\theta = \sin^{-1}\left\{\frac{25}{50}\right\}$$

$$\theta = 30^\circ$$



$$\frac{W}{\sin 105^\circ} = \frac{P}{\sin 120^\circ} \Rightarrow P = 896.575 \text{ N}$$

Alternate Method:

$$\sum M_B = 0 \Rightarrow P \cos 45^\circ \{25\} + P \sin 45^\circ \{25\sqrt{3}\} = 1000 \{25\sqrt{3}\}$$

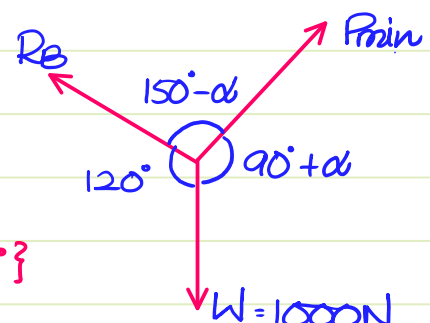
$$P \left\{ \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}} \right\} = 1000\sqrt{3} \quad P = 896.575 \text{ N}$$

→ find the inclination of P w.r.t. horizontal & corresponding value of P if P is Minimum.
If inclination of $P_{\min}$ w.r.t. horizontal is α .

$$\frac{W}{\sin \{150^\circ - \alpha\}} = \frac{P_{\min}}{\sin 120^\circ}$$

$\Rightarrow \sin(150^\circ - \alpha)$ must be Maximum $\{1\}$
 $\{\alpha = 60^\circ\}$

$$P_{\min} = 866.025 \text{ N}$$



2.2. Analysis of Structures

A. Trusses

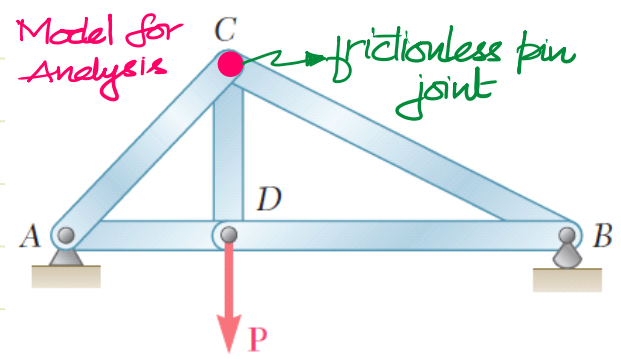
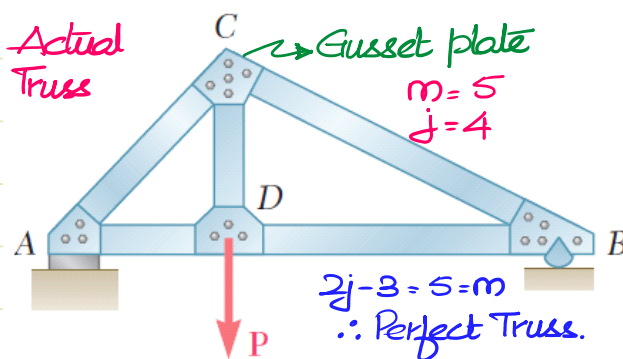
- * Introduction
- * Identification of Support Rx^{ns}.
- * Method of Joints {1M}
- * Method of Sections {1M (or) 2M}
- * Identification of Zero force Members.

B. Frames.

- * Introduction
- * Identification of loadings in Members. {2M}

A. Trusses

A structure formed by joining uniform slender rods in order to carry loads.



An actual truss has Gusset plates over which the slender members are joined by bolting / welding etc. In order to analyse the Gusset plates are replaced by frictionless pin joints.

Perfect Truss: A truss in which the combination of No. of Members & the No. of joints is such that it forms a perfect structure.

Eqⁿ of Perfect Truss:

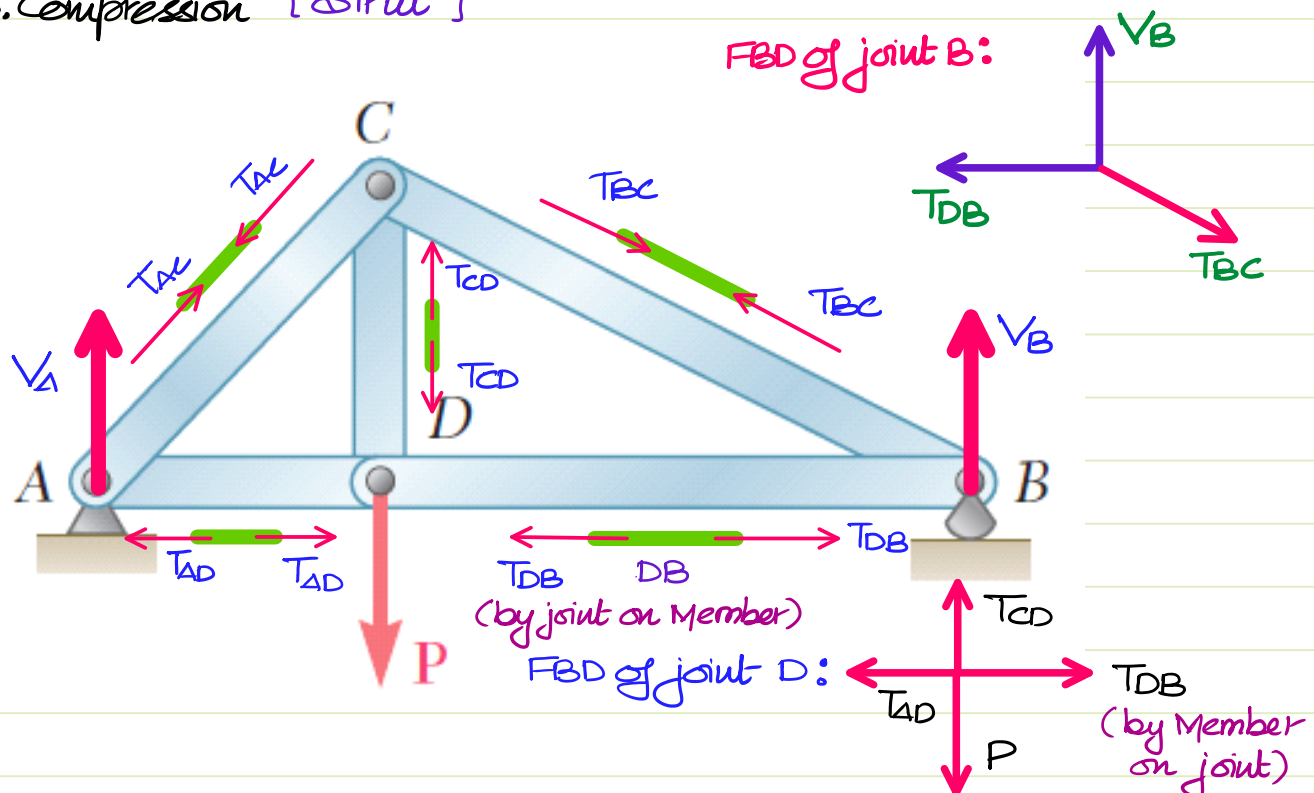
$$m = 2j - 3$$

m : No. of Members
 j : No. of joints.

$m > 2j - 3$ {Truss with Redundant Member}
 $m < 2j - 3$ {Collapsible truss}

Assumptions Required In The Analysis of Trusses:

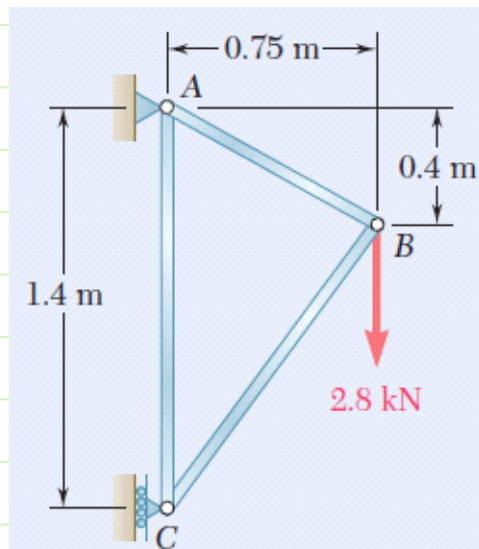
1. The weight of truss members are Neglected.
2. All the joints are frictionless pin joints {No moment reacⁿ.}
3. All the applied loadings & support reac^{ns} act on the joints.
The joints then apply certain forces on the adjacent members which create reacⁿ back on the joints. All these forces together establish eq^m on the joints.
4. Every member is a 2 force Member {all members have axial loads?}
 - A. Tension {Tie / Tie Rod}
 - B. Compression {Strut}



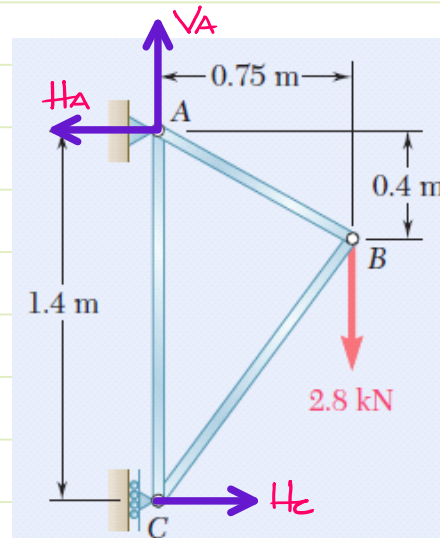
Note: Every member is assumed to be a tensile member.
 If the loading comes out to be Negative, implies that Member is in compression.

Identification of Support Reactions: Consider the entire structure as a single body & establish eqⁿ {disregarding the loadings in the Members}

Q: find the support reac^{ns} in the given truss



Solution:



$$\sum M_A = 0$$

$$H_C \{1.4\} - 2.8 \{0.75\} \\ \Rightarrow H_C = 1.5 \text{ kN.}$$

$$\sum F_x = 0$$

$$H_A = 1.5 \text{ kN.}$$

$$\sum F_y = 0$$

$$V_A = 2.8 \text{ kN.}$$

** Analysis of Trusses By Method of joints:

Choose a joint, draw the FBD by considering applied loadings, support reactions & the reac^{ns} from the Members.

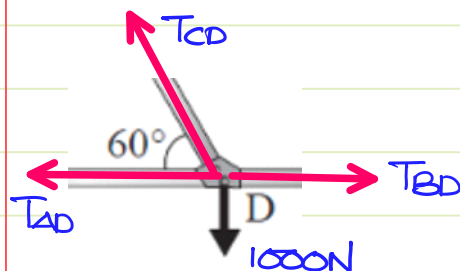
When to use this Method?

1. If a joint has only 2 unknown loadings.
2. If a joint has more than 2 unknown loadings but all the loadings except the one to be found are all collinear.

If we observe the joint "D",
it has 3 unknown loadings
{ T_{AD} , T_{BD} & T_{CD} }

We observe T_{AD} & T_{BD} are
collinear in x axis.

$\therefore \sum F_y = 0$, then only one
unknown acts in this axis.

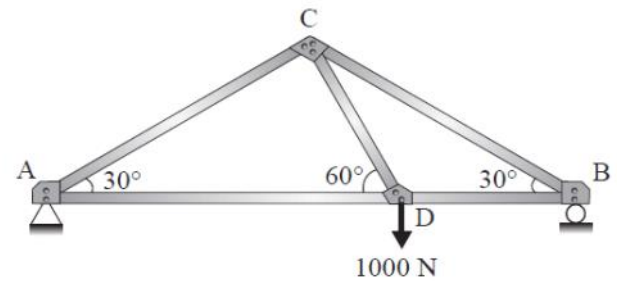


$$\sum F_y = 0$$

$$T_{CD} \sin 60^\circ - 1000 = 0$$

$$T_{CD} = 1154 \text{ N}$$

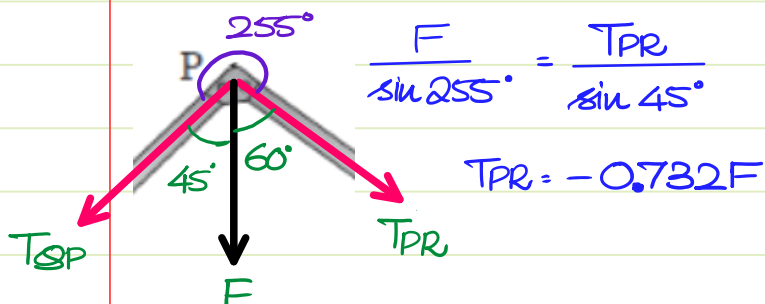
01. The axial force in the bar CD of the simple truss supported and loaded as shown in the figure below is



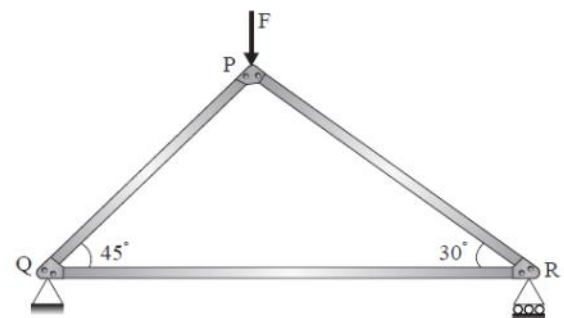
- (a) 1000 N
(b) 1154 N
(c) 1500 N
(d) 750 N

Step 1: MOJ at P to find T_{PR}

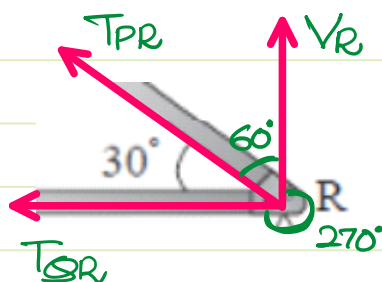
Step 2: MOJ at R to find T_{OR}



04. Consider a truss PQR loaded at P with a force F as shown in the figure. The tension in the member QR is
(GATE - 2008)



- (a) 0.5 F
(b) 0.63 F
(c) 0.73 F
(d) 0.87 F

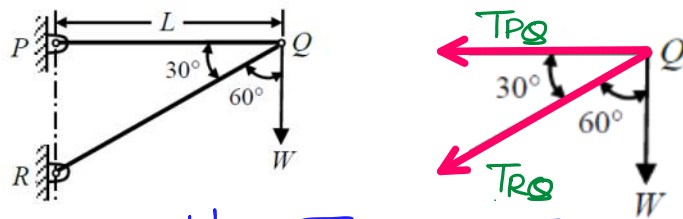


$$\frac{T_{PR}}{\sin 270^\circ} = \frac{T_{OR}}{\sin 60^\circ}$$

$$\frac{-0.732 \cdot \sin 60^\circ \cdot F}{-1} = T_{OR}$$

$$T_{OR} = 0.634 F$$

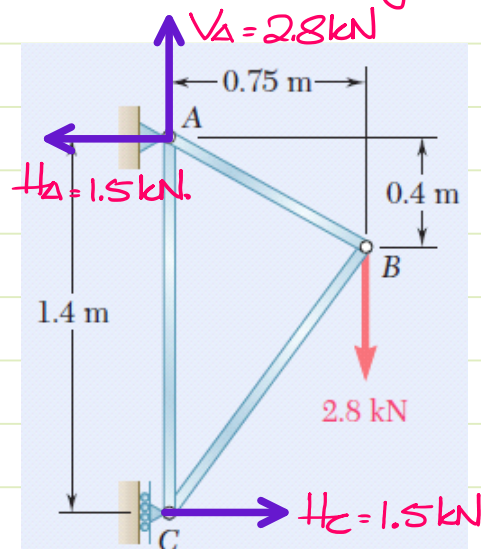
A two-member truss PQR is supporting a load W . The axial forces in members PQ and QR are respectively



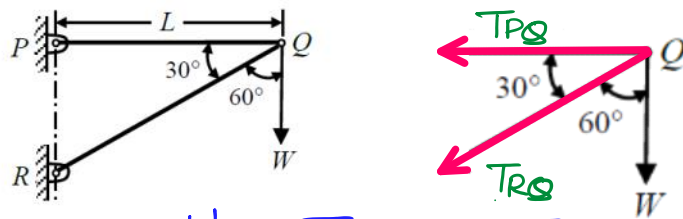
- (A) $2W$ tensile and $\sqrt{3}W$ compressive
 ✓ (B) $\sqrt{3}W$ tensile and $2W$ compressive
 (C) $\sqrt{3}W$ compressive and $2W$ tensile
 (D) $2W$ compressive and $\sqrt{3}W$ tensile

$$\frac{W}{\sin 30^\circ} = \frac{T_{PQ}}{\sin 60^\circ} = \frac{T_{RQ}}{\sin 270^\circ}$$

Home Work : find the loadings in all the Members in the given truss system.



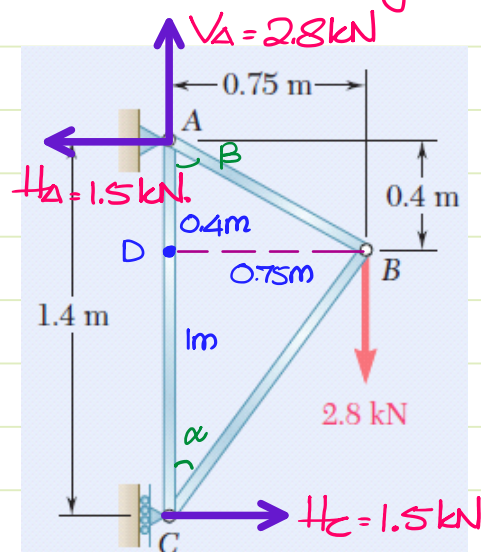
A two-member truss PQR is supporting a load W . The axial forces in members PQ and QR are respectively



- (A) $2W$ tensile and $\sqrt{3}W$ compressive
 (B) $\sqrt{3}W$ tensile and $2W$ compressive
 (C) $\sqrt{3}W$ compressive and $2W$ tensile
 (D) $2W$ compressive and $\sqrt{3}W$ tensile

$$\frac{W}{\sin 30^\circ} = \frac{T_{PQ}}{\sin 60^\circ} = \frac{T_{RQ}}{\sin 270^\circ}$$

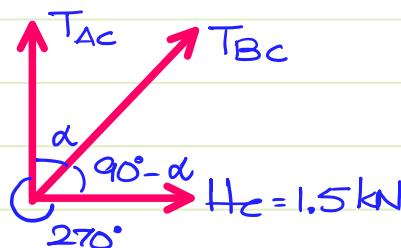
Home Work: find the loadings in all the Members in the given truss system.



$$\tan \alpha = 3/4 \Rightarrow \cos \alpha = 4/5 ; \sin \alpha = 3/5$$

$$\tan \beta = 15/8 \Rightarrow \cos \beta = 8/17 ; \sin \beta = 15/17$$

By Method of joints at "C":

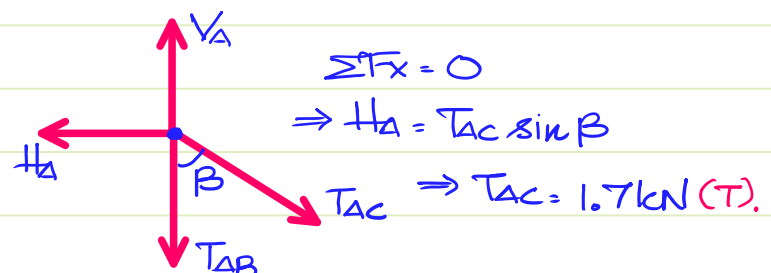


$$\frac{H_C}{\sin \alpha} = \frac{T_{AC}}{\cos \alpha} = \frac{T_{BC}}{\sin 270^\circ}$$

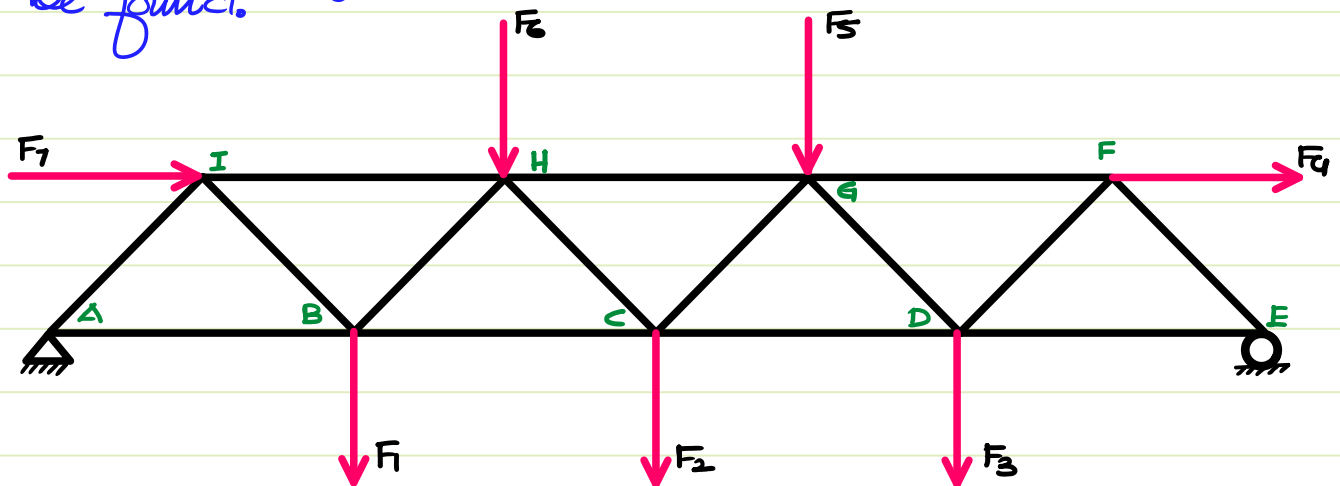
$$\therefore T_{AC} = \frac{1.5}{\tan \alpha} = \frac{1.5 \times 4}{3} = 2 \text{ kN. } \{T\}$$

$$T_{BC} = -2.5 \text{ kN} = 2.5 \text{ kN } \{C\}$$

By Method of Joints @ A:



Analysis of Trusses By Method of Section: This method is preferable when loading in any intermediate member of a truss has to be found.



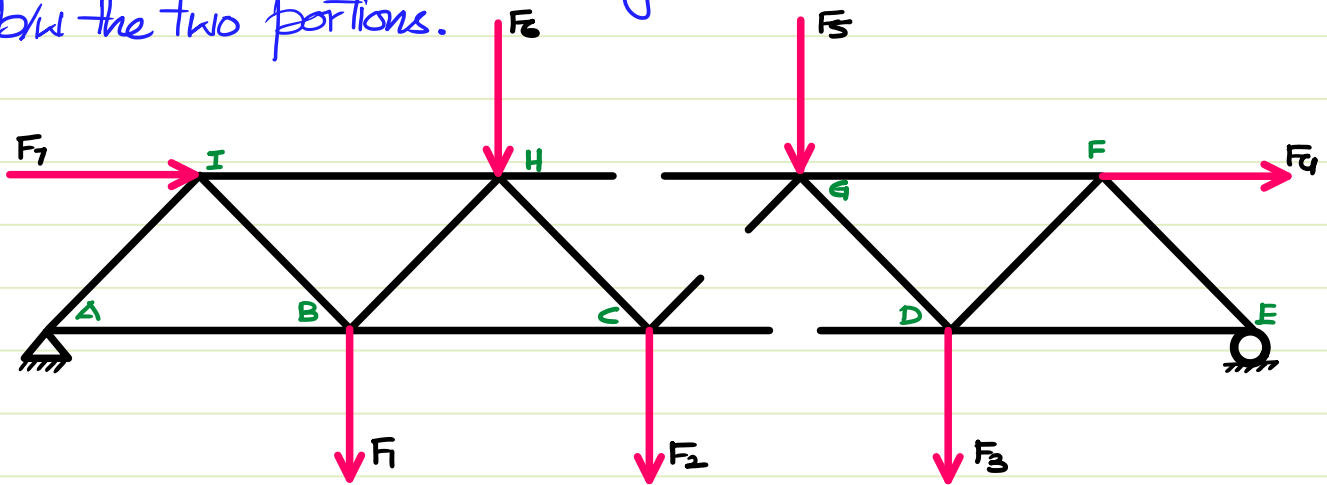
Find the loading in the Member CG by Method of joints.
 Step 1: find the support reacⁿ at E
 Step 2: Use M.O.J at E to find T_{EF} & T_{ED}
 Step 3: Use M.O.J at F to find T_{FG} & T_{DF}
 Step 4: Use M.O.J at D to find T_{DG} & T_{CD}
 Step 5: Use M.O.J at G to find T_{CG} .
 The solution becomes extremely lengthy by the above steps.

$\therefore$ We prefer Method of sections. In this method, eq^m of a portion of Truss is established by considering the forces on the incomplete members, applied loadings & support reac^{ns} present in that portion. We disregard loadings in complete members since they are Internal forces.

Note: Method of sections is based on the idea, if a system is in eq^m, then any portion of the system must also be in equilibrium.

Steps To Be followed in M.O.S.:-

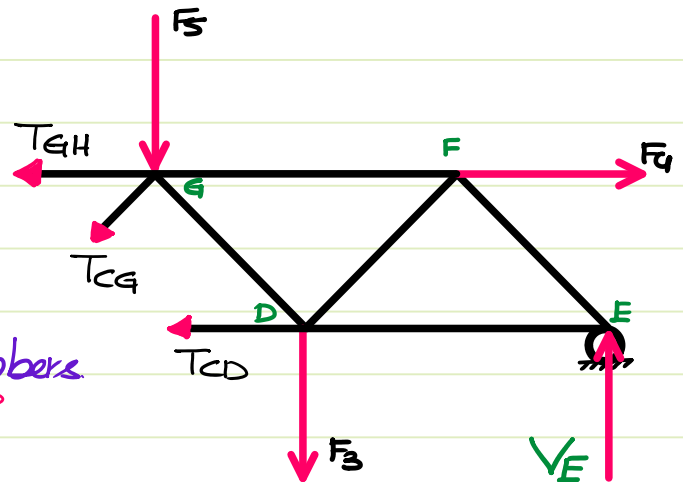
Step 1: Cut the Truss in such a way that there is no connection b/w the two portions.



Ideally cut 3 Members. Always start cutting the Member in which loading has to be identified.

Step 2: Choose a portion & draw its FBD. Establish eqm by considering

1. Support Rx^{ns} {V_E}
2. Applied loadings {F₃, F₄, F₅}
3. Loadings in Incomplete Members {T_{GH}, T_{CG}, T_{CD}}



Step 3: Try to identify the unknown in a single step.

find T_{CG}
observe T_{GH} & T_{CD}
∴ they are ||^{al} in x
∴ $\sum F_y = 0$

find T_{GH}
observe T_{CG} & T_{CD}
∴ they are conc at C
∴ $\sum M_C = 0$

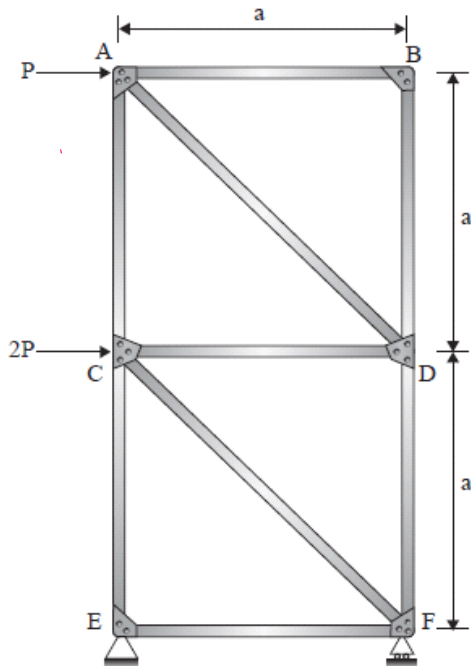
find T_{CD}
observe T_{GH} & T_{CG}
∴ they are conc at G
∴ $\sum M_G = 0$

* While attempting any problem by M.O. Sections ask yourself.

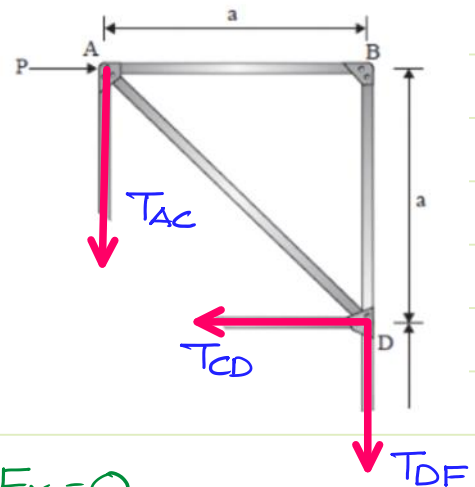
1. Which Members to cut
2. Which side to consider {Reduce finding Support Rx^{ns}}
3. Which eqⁿ to write

1 Mark

02. The force in the member CD of the truss shown in the figure below is



- (a) Zero
(b) $2P$ (Comp)
(c) P (Comp)
(d) P (Tensile)



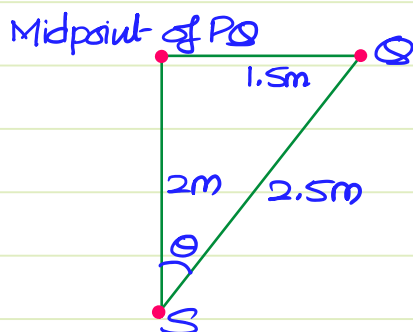
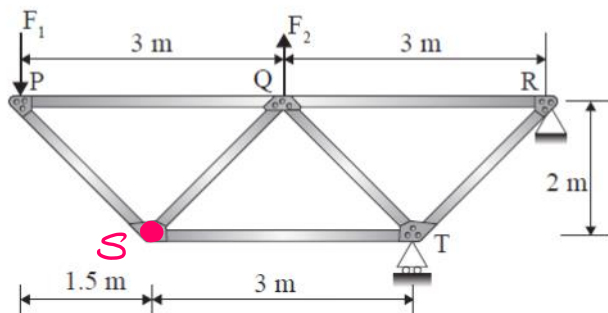
$$\sum F_x = 0$$

$$P - T_{CD} = 0$$

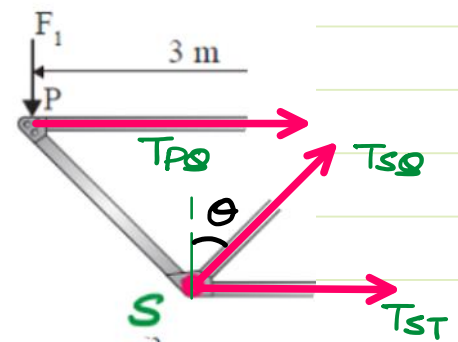
$$T_{CD} = P \text{ (T)}$$

2 Mark

07. For the truss shown in the figure, the forces F_1 and F_2 are 9 kN and 3 kN , respectively. The force (in kN) in the member QS is (GATE-2014)



$$\cos \theta = \frac{4}{5}$$



$$\sum F_y = 0$$

$$-F_1 + T_{SQ} \cos \theta = 0$$

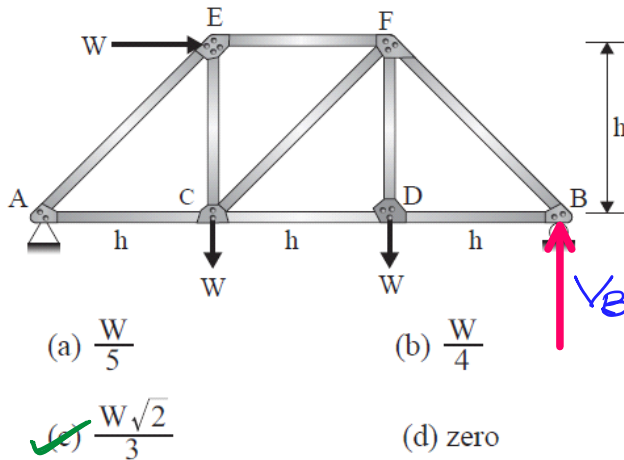
$$T_{SQ} = \frac{9 \text{ kN}}{\cos \theta}$$

$$T_{SQ} = \frac{9 \text{ kN} \times 5}{4}$$

$$T_{SQ} = 11.25 \text{ kN (T)}$$

2 Mark

08. The force in the member CF of the truss shown in the figure below is



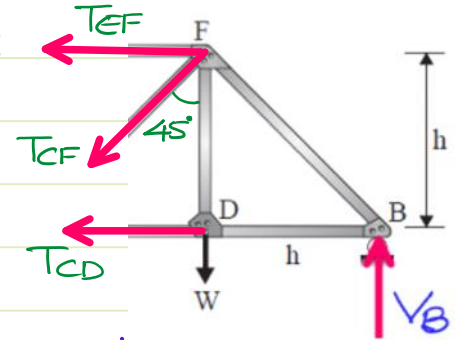
Step 1: find V_B

$\sum M_A = 0$ for entire structure

$$-Wh - Wh - W(2h) + V_B(3h) = 0$$

$$V_B = \frac{4W}{3}$$

Step 2:

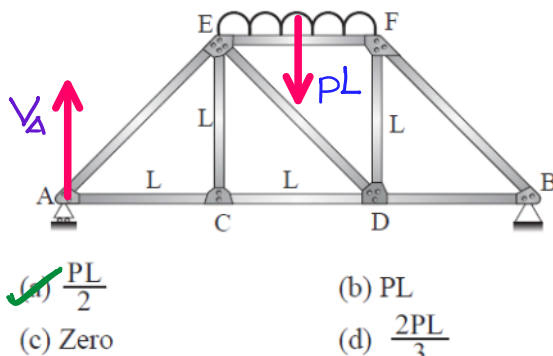


$$\sum F_y = 0 \Rightarrow -T_{CF} \cos 45^\circ - W + \frac{4W}{3} = 0$$

$$T_{CF} = \frac{W\sqrt{2}}{3} \text{ (T)}$$

2 Mark

05. A truss consists of horizontal members and vertical members having length L each. The members AE, DE and BF are inclined at 45° to the horizontal. For the uniformly distributed load P per unit length on the member EF of the truss shown in the figure, the force in the member CD is (GATE - 2003)



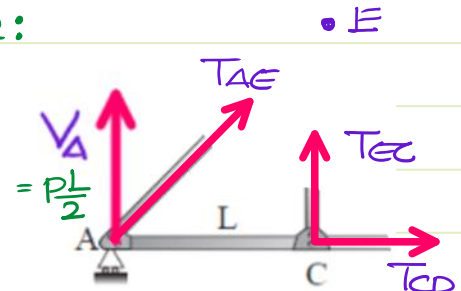
Step 1: find V_A

$$\sum M_B = 0$$

$$-V_A \{3L\} + PL \left\{ \frac{3L}{2} \right\} = 0$$

$$\Rightarrow V_A = \frac{PL}{2}$$

Step 2:



$$\sum M_E = 0$$

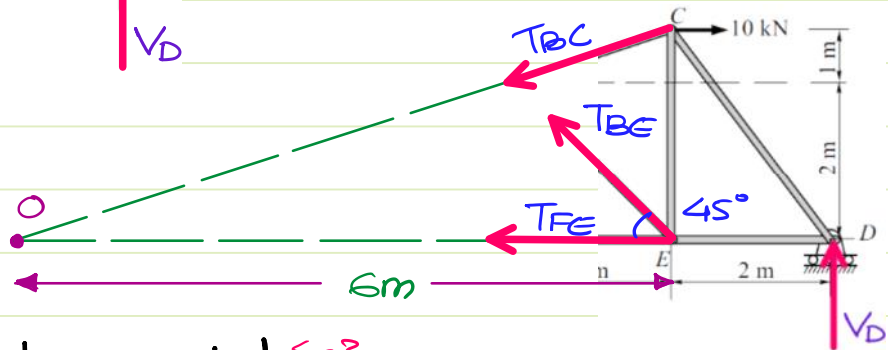
$$-\frac{PL}{2} \{L\} + T_{CD} \{L\} = 0$$

$$T_{CD} = \frac{PL}{2} \text{ (T)}$$

find the loading in the member BE

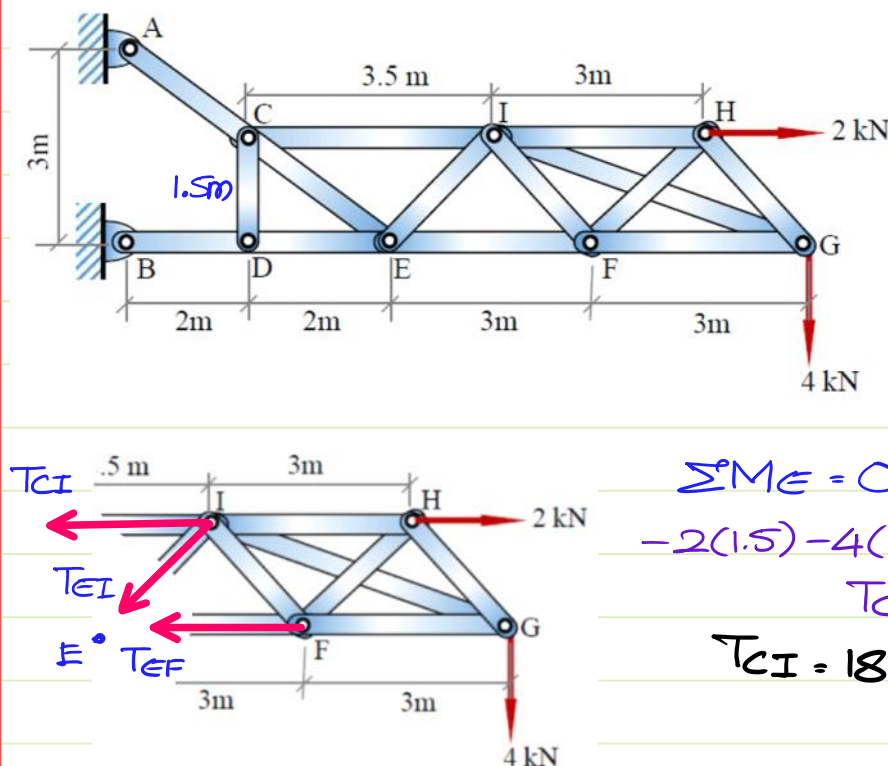
$\Sigma^p M_A = 0$ for the net structure

$$V_D = 15 \text{ kN.}$$


$$+ T_{BE} \sin 45 \{6\} = 0$$

$$T_{BE} = -21.21 \text{ kN} = 21.21 \text{ kN} \{C\}$$

A structure, along with the loads applied on it, is shown in the figure. Self-weight of all the members is negligible and all the pin joints are friction-less. AE is a single member that contains pin C. Likewise, BE is a single member that contains pin D. Members GI and FH are overlapping rigid members. The magnitude of the force carried by member CI is kN (in integer).



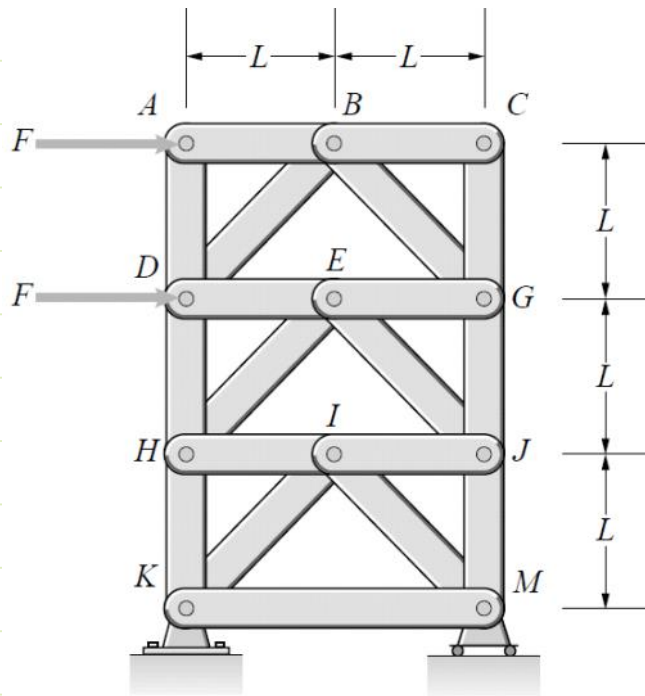
$$\sum^p M_i = 0$$

$$-2(1.5) - 4(6) + T_{CI}(1.5) = 0$$

$$T_{CI} = 18 \text{ kN}$$

$$T_{CI} = 18 \text{ kN} \{T\}$$

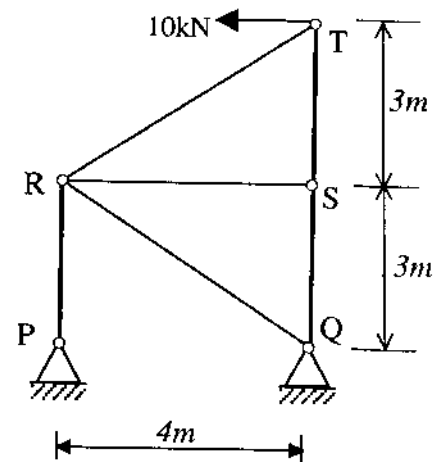
2 Mark



find the loading in the Member DH.

1 Mark

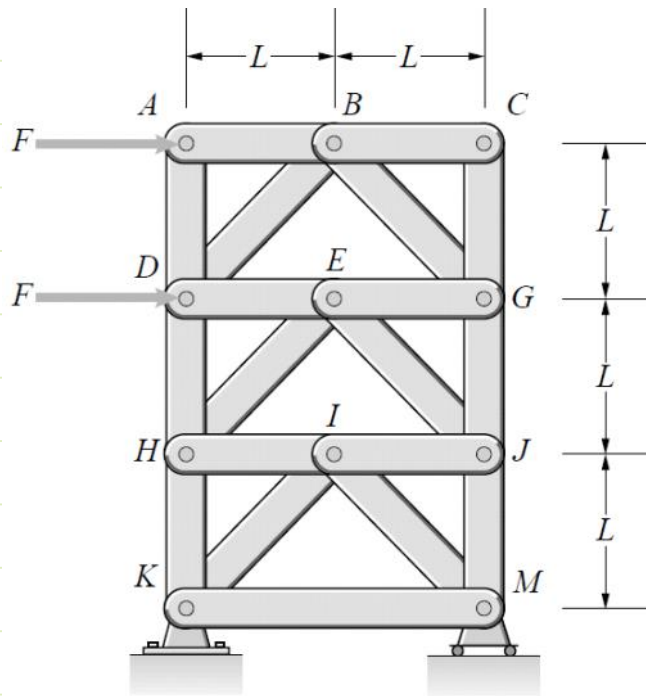
The force in the member PR for the truss shown is



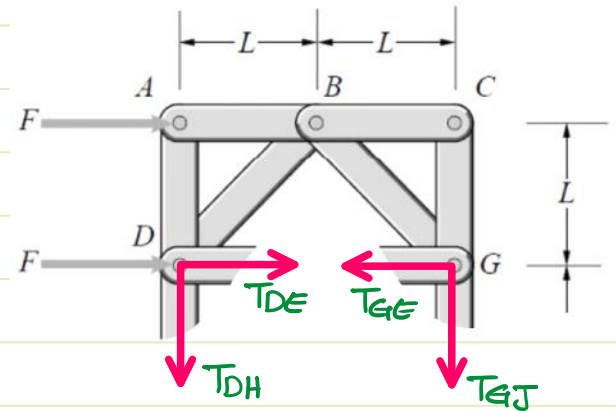
- (A) 12 kN Tensile
- (C) 15 kN Tensile

- (B) 12 kN Compressive
- (D) 15 kN Compressive

2 Mark



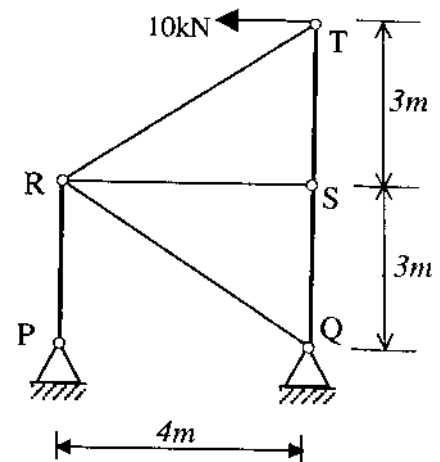
find the loading in the Member DH.



$$\begin{aligned}\sum M_G &= 0 \\ -F\{L\} + T_{DH}\{2L\} &= 0 \\ \Rightarrow T_{DH} &= F/2 \text{ (T)}\end{aligned}$$

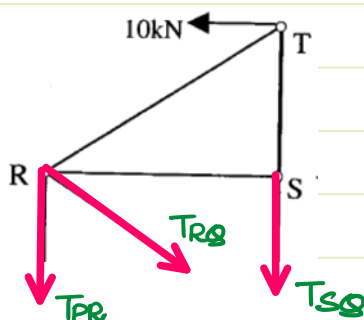
1 Mark

The force in the member PR for the truss shown is



- (A) 12 kN Tensile
(C) 15 kN Tensile

- (B) 12 kN Compressive
(D) 15 kN Compressive



$$\begin{aligned}\sum M_Q &= 0 \\ T_{PR}\{4\} + 10\{6\} &= 0 \\ T_{PR} &= -15 \text{ kN} \\ &= 15 \text{ kN (Compression)}\end{aligned}$$

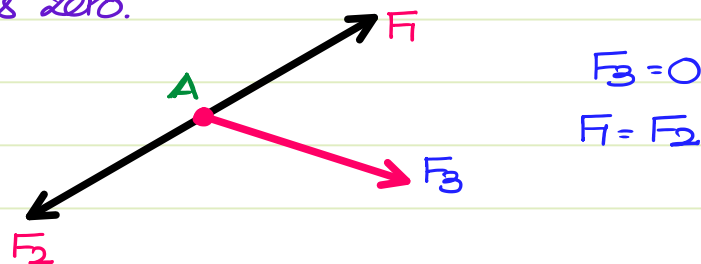
* Zero force Members: The members in a truss system which carry zero loading are known as Zero force Members.

Why do we include zero force Members?

1. Including zero force Members reduces the length of comp. members {struts?}. This reduces their chances of buckling.
2. In case of variable loadings, zero force members may carry load.

** How to identify Zero force Members?

1. If there is a joint having a total of 3 forces, such that 2 of these forces are collinear & the 3rd force {in the form of loading in Member} is Non collinear, then the 3rd force is zero.



2. If a joint has 2 Non collinear forces {both are in the form of Member loadings}, then both Members are Zero force Members.



3. Identification of loadings in individual Members to find Zero force Members.

Note: Once a zero force Member is identified, then virtually remove that Member & re-examine to find further zero force Members.